Cross-step interference, directional curvature, and gradient descent

1 Introduction

Training large language models (LLMs) relies on adaptive optimization, momentum, gradient clipping, learning-rate schedules and other heuristics to stabilize learning and mitigate phenomena such as loss spikes and oscillatory updates. Yet, despite their effectiveness, it is not fully understood why these mechanisms work, and under what conditions they fail.

To address this gap, we identify a persistent behaviour of transition between two distinct learning regimes. Early in training, successive gradients tend to align, allowing updates to accumulate constructively as the loss decreases. Later, gradient directions reverse more frequently, causing successive updates to partially cancel one another. Despite this increased cancellation, the loss can continue to decrease.

Two complementary intuitions help explain this behavior. From the perspective of representation learning, each new mini-batch updates parameters that also encode previously learned structure. Depending on its direction, an update may reinforce existing representations or partially undo them. From the perspective of optimization geometry, a step may overshoot the bottom of a loss valley, prompting subsequent steps to reverse direction and partially retrace the path. Both mechanisms can give rise to oscillatory parameter updates even as optimization continues to make progress.

We formalize these dynamics using identities for stochastic gradient descent (SGD), relating the alignment of consecutive gradients to curvature and gradient noise, and cumulative update cancellation to net parameter displacement. Building on these relations, we introduce scalable diagnostics for quantifying these dynamics and tracking the transition between constructive and oscillatory regimes during large-scale model training.

2 Parameter displacement under stochastic gradient descent

Let $\theta_t \in \mathbb{R}^p$ denote the parameters after update t, and let g_t denote a mini-batch gradient evaluated at θ_{t-1}. We consider constant-step SGD with learning rate $\eta > 0$, without momentum, weight decay, or clipping

$$\theta_t = \theta_{t-1} - \eta g_t. \tag{1}$$

After T updates, summing Eq. (1) gives the displacement from initialization,

$$\Delta_T := \theta_T - \theta_0 = -\eta \sum_{t=1}^{T} g_t. \tag{2}$$

Expanding its squared norm separates individual gradient magnitudes from their temporal interactions

$$\|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 + 2\eta^2 \sum_{1 \le u < t \le T} g_u^\top g_t. \tag{3}$$

The gradient-energy terms $\|g_t\|^2$ are nonnegative. Each pairwise inner product $g_u^\top g_t$ can be positive, zero, or negative: aligned gradients reinforce their contribution to the sum, whereas opposing gradients cancel part of it. To practically measure the alignement of nonzero consecutive gradients, define their cosine similarity as

$$\rho_t := \frac{g_{t-1}^\top g_t}{\|g_{t-1}\| \, \|g_t\|}, \qquad t = 2, \dots, T. \tag{4}$$

Separating adjacent pairs from the remaining interactions yields

$$\begin{aligned} \|\Delta_T\|^2 = \eta^2 \sum_{t=1}^{T} \|g_t\|^2 \\ + 2\eta^2 \underbrace{\sum_{t=2}^{T} \|g_{t-1}\| \, \|g_t\| \, \rho_t}_{\text{lag-one interaction}} \\ + 2\eta^2 \underbrace{\sum_{t=3}^{T} \sum_{u=1}^{t-2} g_u^\top g_t}_{\text{interaction beyond lag one}}. \end{aligned} \tag{5}$$

Negative ρ_t contributes cancellation between consecutive updates. However, longer-lag interactions can reinforce or offset these negative contributions. Thus, adjacent anti-alignment diagnoses a local interaction, while the complete double sum determines longer-term cumulative interaction.

2.1 Curvature, gradient noise, and consecutive alignment

For mini-batch SGD, we explicitly incorporate stochastic gradient noise into the evolution of consecutive gradients and examine its contribution to their directional alignment. Let $\mathcal{L}$ denote the full training objective. We write a stochastic gradient as

$$g_t = \nabla \mathcal{L}(\theta_{t-1}) + \xi_t, \tag{6}$$

where ξ_t is the gradient estimation error. Suppose $\mathcal{L}$ is twice continuously differentiable along the step under consideration, and define $H_{t-1} := \nabla^2 \mathcal{L}(\theta_{t-2})$. Expanding the full gradient around θ_{t-2} and substituting the preceding SGD step gives

$$g_t = (I - \eta H_{t-1}) g_{t-1} + (\xi_t - \xi_{t-1}) + r_t, \tag{7}$$

where r_t is the Taylor remainder. Neglecting r_t recovers the first-order approximation. Taking the inner product with g_{t-1} then gives

$$\rho_t \approx \frac{\underbrace{\|g_{t-1}\|^2}_{\text{Gradient magnitude}} - \underbrace{\eta g_{t-1}^\top H_{t-1} g_{t-1}}_{\text{Curvature contribution}} + \underbrace{g_{t-1}^\top (\xi_t - \xi_{t-1})}_{\text{Noise contribution}}}{\|g_{t-1}\| \|g_t\|}. \tag{8}$$

The numerator contains the preceding gradient energy, the curvature contribution, and the change in gradient noise. Positive curvature along the preceding gradient direction reduces its alignment with the next gradient. The noise difference can reinforce or oppose this effect on an individual step.

We normalize the curvature and noise terms by the preceding gradient energy:

$$\kappa_t := \frac{g_{t-1}^\top H_{t-1} g_{t-1}}{\|g_{t-1}\|^2}, \qquad \varepsilon_t := \frac{g_{t-1}^\top (\xi_t - \xi_{t-1})}{\|g_{t-1}\|^2}. \tag{9}$$

With these definitions, Eq. (8) becomes

$$\boxed{\rho_t \approx \frac{\|g_{t-1}\|}{\|g_t\|}(1 - \eta\kappa_t + \varepsilon_t).} \tag{10}$$

The norm ratio is positive, so the first-order sign prediction depends on $1 - \eta\kappa_t + \varepsilon_t$ (Table 1).

Table 1: Consecutive-gradient alignment predicted by the first-order model. The Taylor remainder is neglected; these are not exact sign equivalences for an arbitrary nonlinear step.

Alignment	Cosine sign	First-order condition
Positive	$\rho_t > 0$	$\eta\kappa_t < 1 + \varepsilon_t$
Orthogonal	$\rho_t = 0$	$\eta\kappa_t = 1 + \varepsilon_t$
Negative	$\rho_t < 0$	$\eta\kappa_t > 1 + \varepsilon_t$

2.1.1 Directional curvature

The quantity κ_t is a Hessian Rayleigh quotient along the preceding stochastic gradient. If $H_{t-1} v_i = \lambda_i v_i$ in an orthonormal eigenbasis and $g_{t-1} = \sum_i a_i v_i$, then

$$\kappa_t = \frac{\sum_i \lambda_i a_i^2}{\sum_i a_i^2}. \tag{11}$$

Thus, the gradient selects an eigenvalue-weighted average rather than the maximum eigenvalue. A sharp direction can exist without contributing much to the current gradient. This distinction is important when comparing alignment with the largest Hessian eigenvalue.

Rearranging Eq. (10) motivates the observable quantity

$$\eta\widehat{\kappa}_t := 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} + \varepsilon_t. \tag{12}$$

Equation (7) gives $\widehat{\kappa}_t = \kappa_t - g_{t-1}^\top r_t / (\eta \|g_{t-1}\|^2)$. Consequently, it approximates the pointwise directional curvature when the Taylor remainder is small. For mini-batch gradients, the noise term requires additional information and cannot be recovered from adjacent cosine and norms alone.

For full-batch GD, $\xi_t = 0$ and hence $\varepsilon_t = 0$. The recorded gradient-only quantity is then exactly

$$\eta\widehat{\kappa}_t = 1 - \rho_t \frac{\|g_t\|}{\|g_{t-1}\|} = -\frac{g_{t-1}^\top (g_t - g_{t-1})}{\|g_{t-1}\|^2}. \tag{13}$$

It measures the change in the gradient projected onto its preceding direction.

2.1.2 Noise contribution

Under fresh, conditionally unbiased mini-batch sampling, we have

$$\mathbb{E}[g_{t-1}^{\top}\xi_t] = 0. \tag{14}$$

But because g_{t-1} already contains the previous noise ξ_{t-1},

$$\mathbb{E}[g_{t-1}^{\top}\xi_{t-1}] = \mathbb{E}[\|\xi_{t-1}\|^2]. \tag{15}$$

Subtracting these two expectations yields

$$\mathbb{E}[g_{t-1}^{\top}(\xi_t - \xi_{t-1})] = -\mathbb{E}[\|\xi_{t-1}\|^2] \le 0. \tag{16}$$

Even unbiased noise therefore produces a nonpositive expected contribution to the unnormalized adjacent inner product. It suggests a systematic mechanism through which stochastic gradient noise can reduce successive-gradient alignment, independently of the local directional curvature.

2.2 Normalized displacement and cumulative interference

Adjacent alignment concerns one pair of steps. To characterize accumulation over the complete trajectory, we compare the sum of gradient lengths with the length of their sum

$$A_T := \sum_{t=1}^{T} \|g_t\|, \qquad B_T := \left\| \sum_{t=1}^{T} g_t \right\|. \tag{17}$$

For Eq. (1), ηA_T is the discrete path length and $\eta B_T = \|\Delta_T\|$ is its endpoint displacement. Provided $A_T > 0$, we define

$$\boxed{R_T := \frac{B_T}{A_T} = \frac{\|\Delta_T\|}{\eta \sum_{t=1}^{T} \|g_t\|} \in [0, 1].} \tag{18}$$

The numerator B_T is the net displacement; the denominator A_T is the total length of all SGD steps. This normalized displacement R_T expresses how much of the accumulated step length remains as net motion of parameters after the gradient directions reinforce or cancel one another.

2.3 Full-batch GD evidence on CIFAR-10

We examine the small CIFAR-10 squared-loss setting used in the released edge-of-stability experiments [1, 2]. We use the first 5,000 CIFAR-10 training images and a fully connected 3072–200–200–10 network with tanh hidden activations and biases (656,810 parameters). Four GD runs share the same frozen initialization from seed zero and use learning rates $\eta \in \{0.004, 2/300, 0.01, 0.02\}$. Each run completes 20,000 updates with no momentum, weight decay, clipping, augmentation, learning-rate schedule, or injected gradient noise. We record loss and raw-gradient diagnostics at every update. The largest Hessian eigenvalue is estimated on all 5,000 examples at initialization and every 100 updates. Figure 1 shows all recorded values.

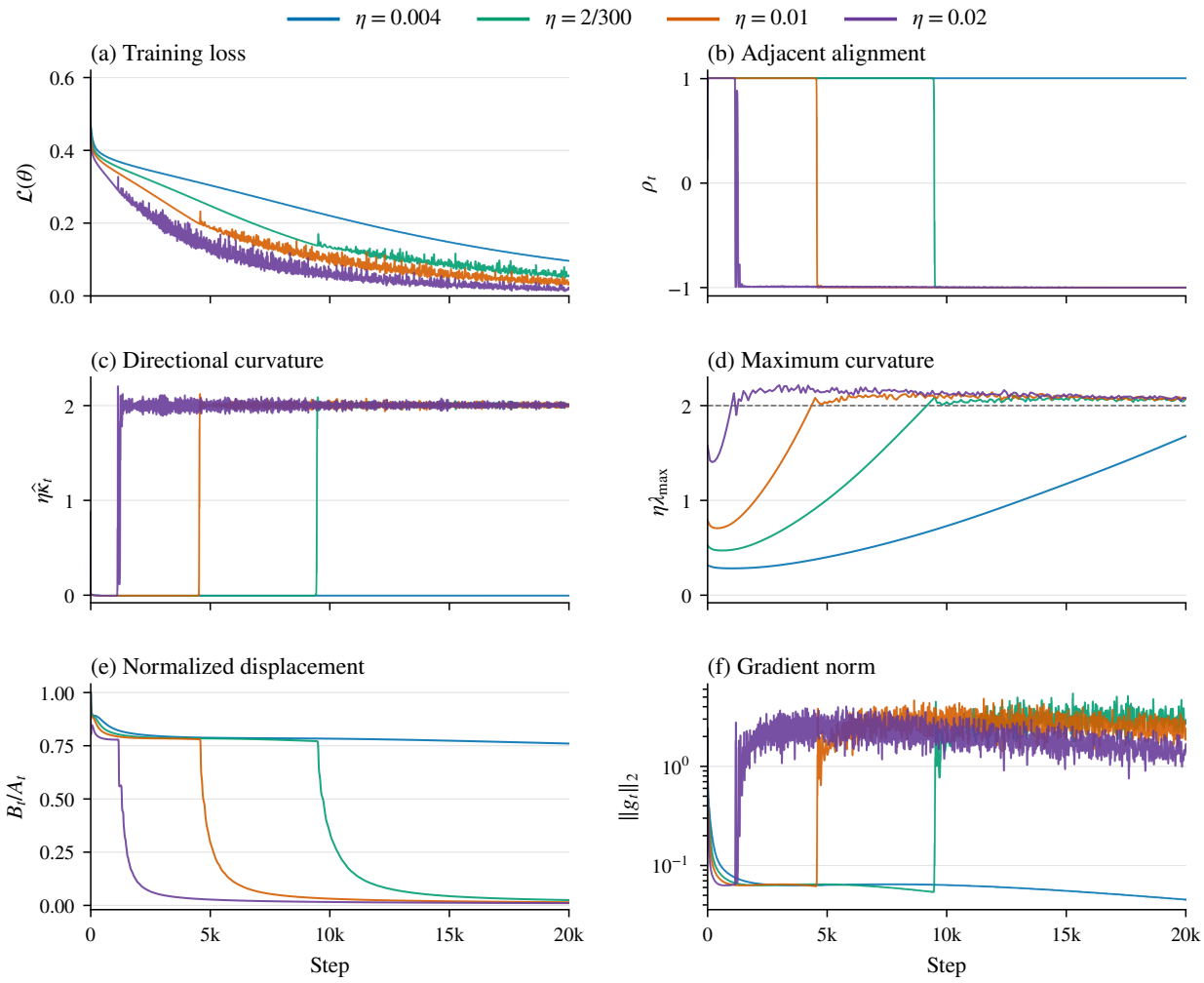


Figure 1: Full-batch GD on CIFAR-10 with four matched learning rates. (a) Training loss; (b) consecutive-gradient cosine; (c) the gradient-derived quantity $\eta\widehat{\kappa}_t = 1-\rho_t \|g_t\| / \|g_{t-1}\|$; (d) the scaled largest algebraic Hessian eigenvalue, with the dashed reference at two; (e) full-prefix normalized displacement B_t/A_t; and (f) raw-gradient norm on log Y. All curves are unsmoothed. Hessians are measured every 100 updates; the other diagnostics are measured every update. Larger learning rates enter the alternating-gradient regime earlier. Loss decreases over the horizon while cumulative cancellation and oscillatory gradient magnitudes increase.

2.4 Alignment reversal and the spectral regime

Figure 1 shows that larger learning rates bring gradient reversal forward. At $\eta = 0.004$, ρ_t stays near one and $\eta\widehat{\kappa}_t$ near zero, even though $\eta\lambda_{\max}$ reaches 1.68. At the three larger rates, ρ_t approaches -1, $\eta\widehat{\kappa}_t$ fluctuates near two, and $\eta\lambda_{\max}$ settles near 2.07. The value $\eta\widehat{\kappa}_t \approx 2$ follows directly from Eq. (13) when consecutive gradients have opposite directions and similar norms; this agreement does not make it a general estimator of maximum curvature. After reversal, gradient norms grow and fluctuate, while B_T/A_T falls to 0.0105–0.0250, compared with 0.7602 at the smallest rate, indicating stronger cancellation relative to path length. Nevertheless, training loss continues to decrease, and final loss falls from 0.0969 to 0.0195 as η increases from 0.004 to 0.02. In this matched, single-initialization comparison, stronger cancellation therefore coexists with greater loss reduction.

TODO: an alignment-guided learning-rate intervention. Test a schedule that halves the learning rate when consecutive-gradient cosine becomes negative, specifying the trigger and repeat rule before execution. Compare loss reduction at a matched update budget with fixed-rate controls. The intervention may restore positive alignment without improving convergence; these two outcomes must be evaluated separately.

References

[1] J. M. Cohen, S. Kaur, Y. Li, J. Z. Kolter, and A. Talwalkar. Gradient descent on neural networks typically occurs at the edge of stability. *International Conference on Learning Representations*, 2021. `https://arxiv.org/abs/2103.00065`.

[2] J. M. Cohen and collaborators. Released implementation of the edge-of-stability experiments, `github` branch. `https://github.com/locuslab/edge-of-stability/tree/github`.